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India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

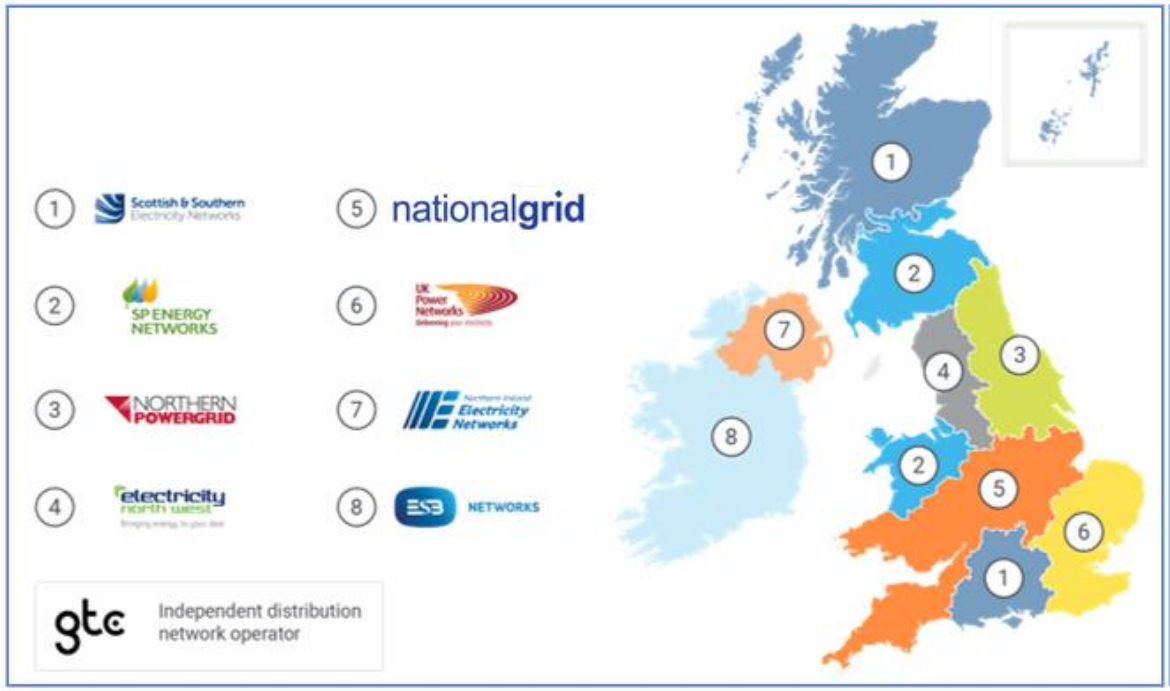
Session : DIGITALIZATION ROADMAPS FOR UTILITIES

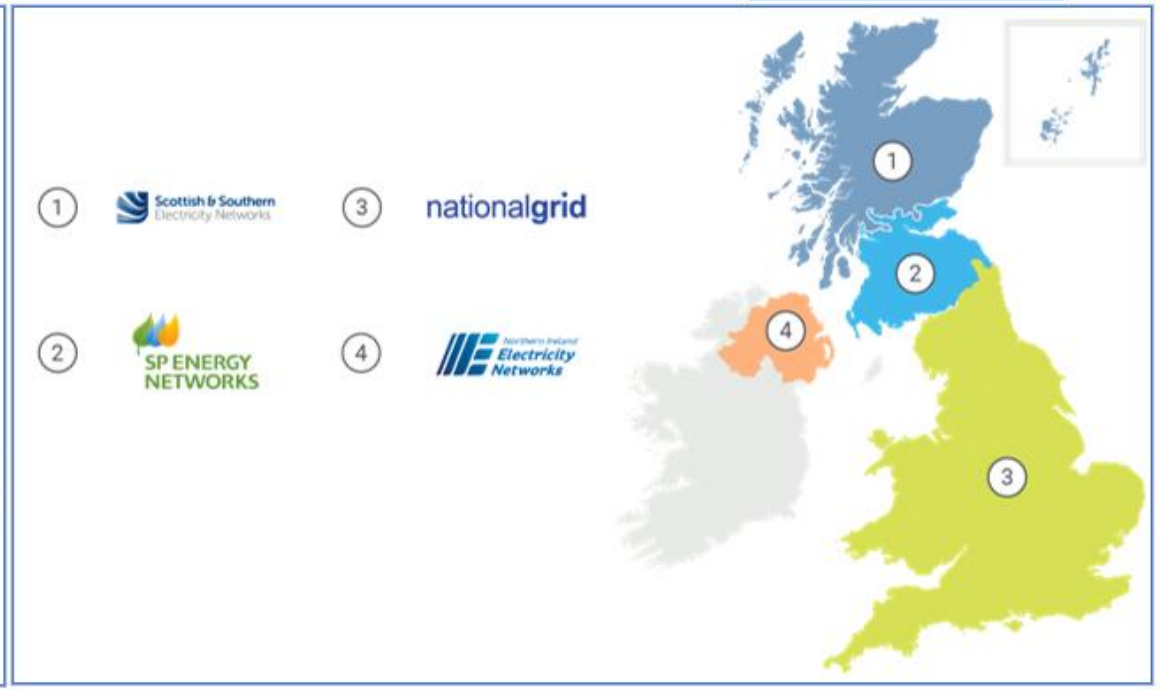
Decarbonization and Digitalization of the grid (The UK's perspective)

Presented By

Dr Avinash Aithal, *Strategic Programme Director, Energy Networks Association UK*

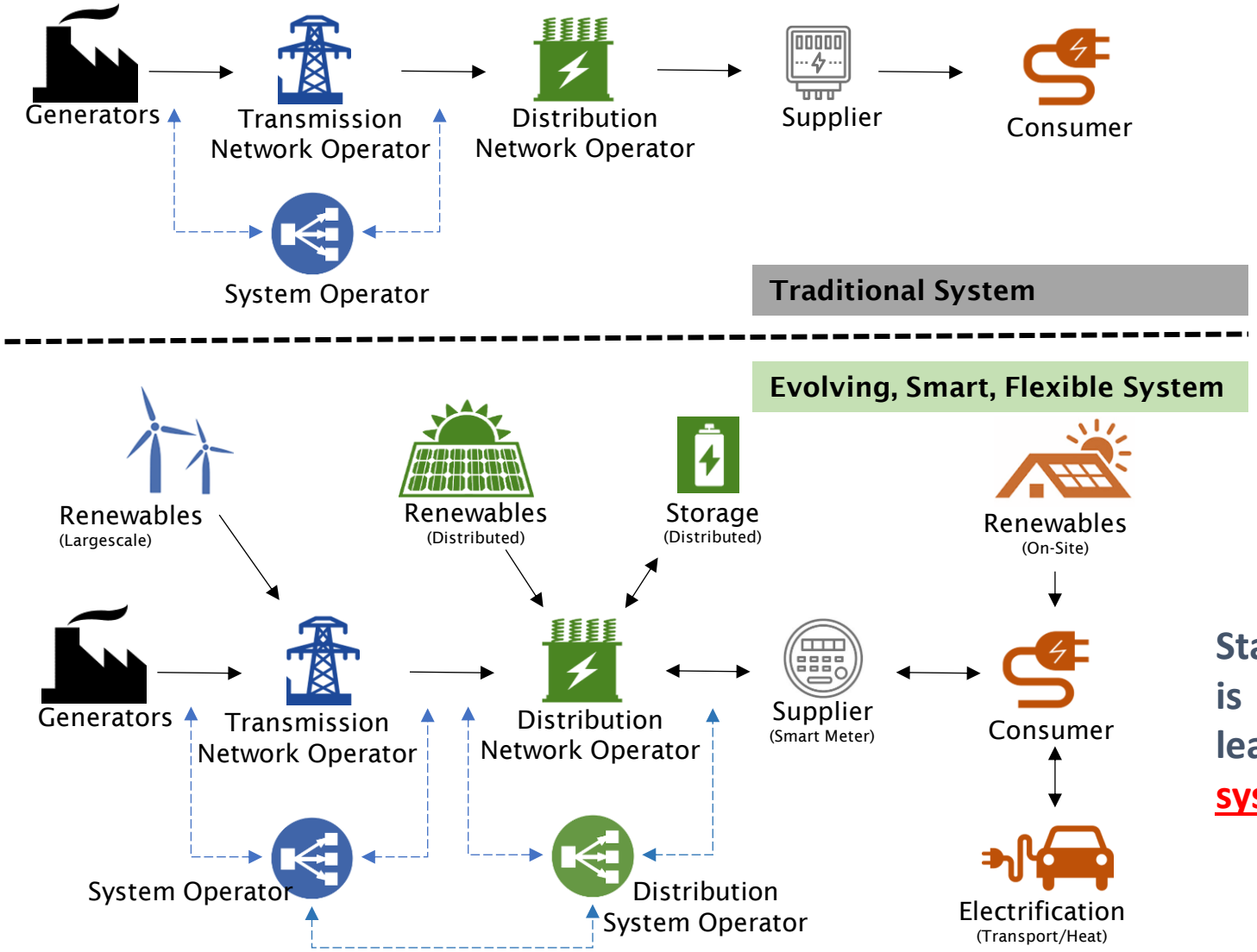
Introduction to ENA





- Construct, operate, maintain and manage network infrastructure in the UK
- Responsible for getting electricity from generators down to customers
- DNOs and TO are privately owned monopoly businesses

The UK's transition to Smart Grid - Open Networks



Started in 2017, the Open Networks programme is working with the networks and industry to lead the transition to a smart and flexible energy system that will enable net zero.

Developing capabilities for Distribution System Operation

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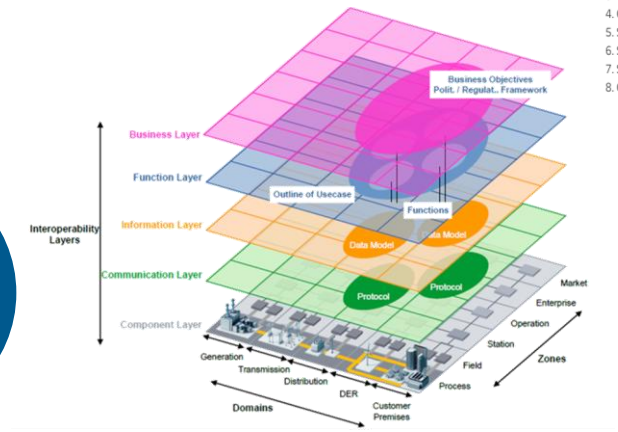
Ideation and vision	Inception and formulation of solutions	Development and implementation	Market proliferation and standardisation	
2017-18	2019-20	2021-22	2023-25	Beyond

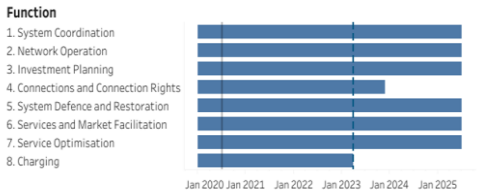
DSO Definition & Functions

Principle of neutral market facilitation

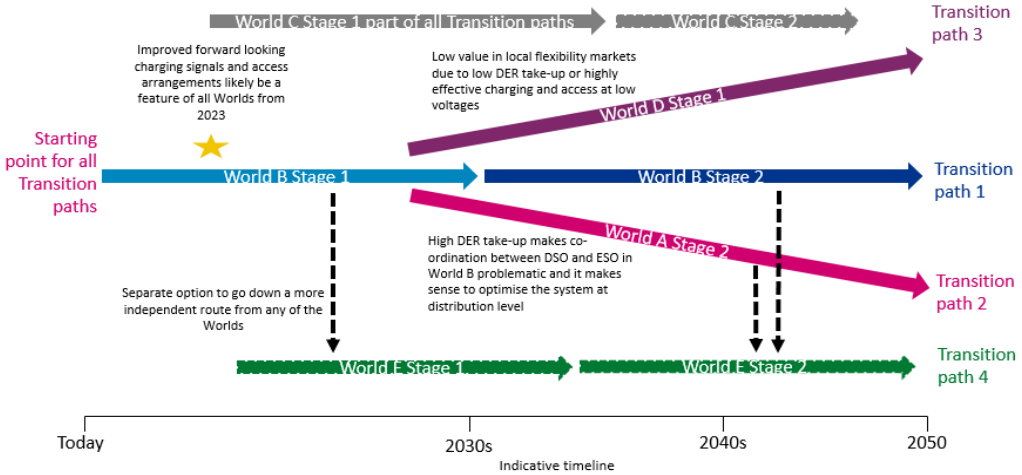
Commercial Principles



Smart grid architecture model analysis



DSO Implementation Plan



Established concept and functions of Distribution System Operation (DSO), including world's first local flexibility markets, built from 'Proof of Concept' into 'Business as Usual'

Flexibility Markets in Great Britain

- **Flexibility is about renumerating a change in where or when electricity is consumed or generated**

Explicit Flexibility

Ancillary Service
- Balancing mechanism
- Frequency response
- Operating reserve
- Black start
- Inertia
- Short circuit
- Reactive power

Capacity Market
- T-1
- T-4

National Markets

Congestion Management
- Peak Reduction
- Scheduled Utilisation
- Operational utilisation
- Operational utilisation +variable availability
- Operational utilisation +Scheduled availability

Local Markets

Regional Development programme (RDPs)

Secondary trading

Implicit Flexibility

Locational Pricing

Time of Use

Retail Markets

National Pricing

Zonal Pricing

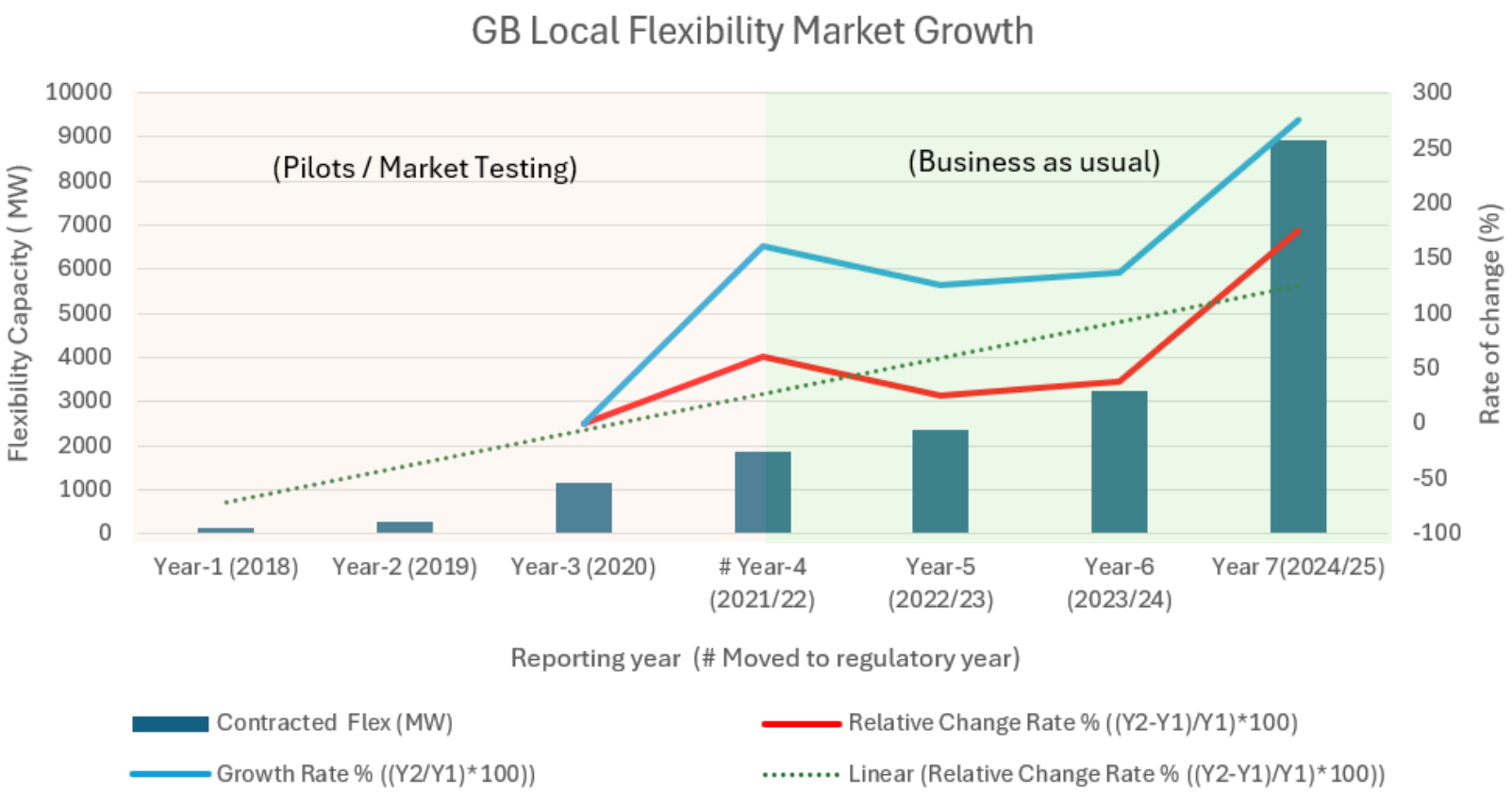
Nodal Pricing

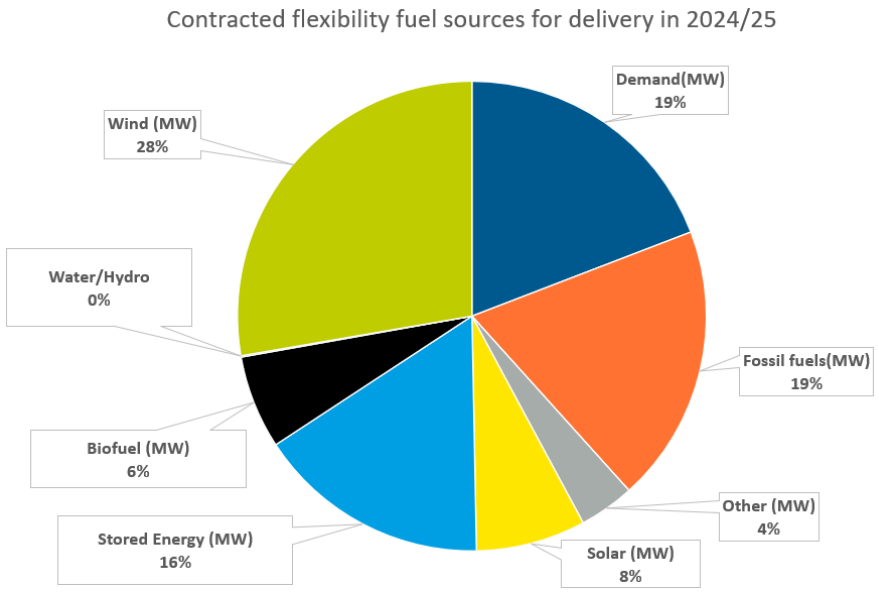
Wholesale Markets

Peer-to-Peer trading

- Open Networks is facilitating the development of local markets and looking at their interaction with national markets.
- GB energy regulator is leading reforms to improve implicit (price driven) flexibility.

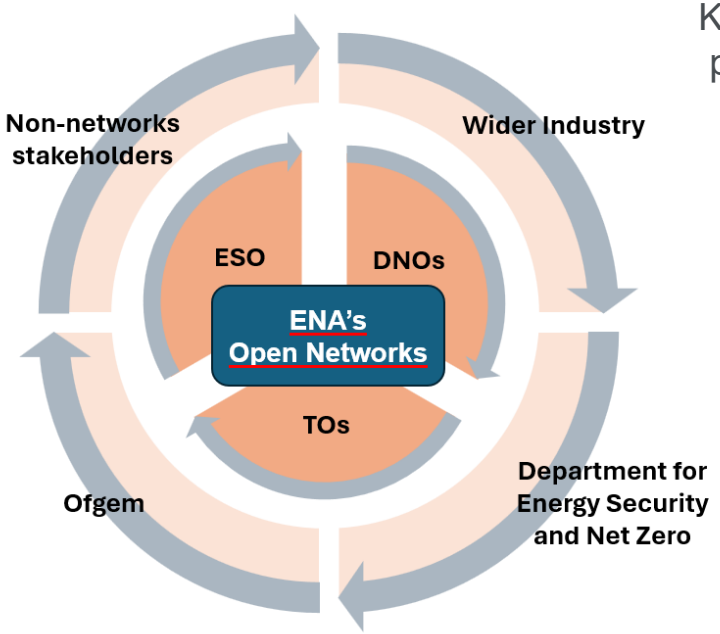
Flexibility market growth

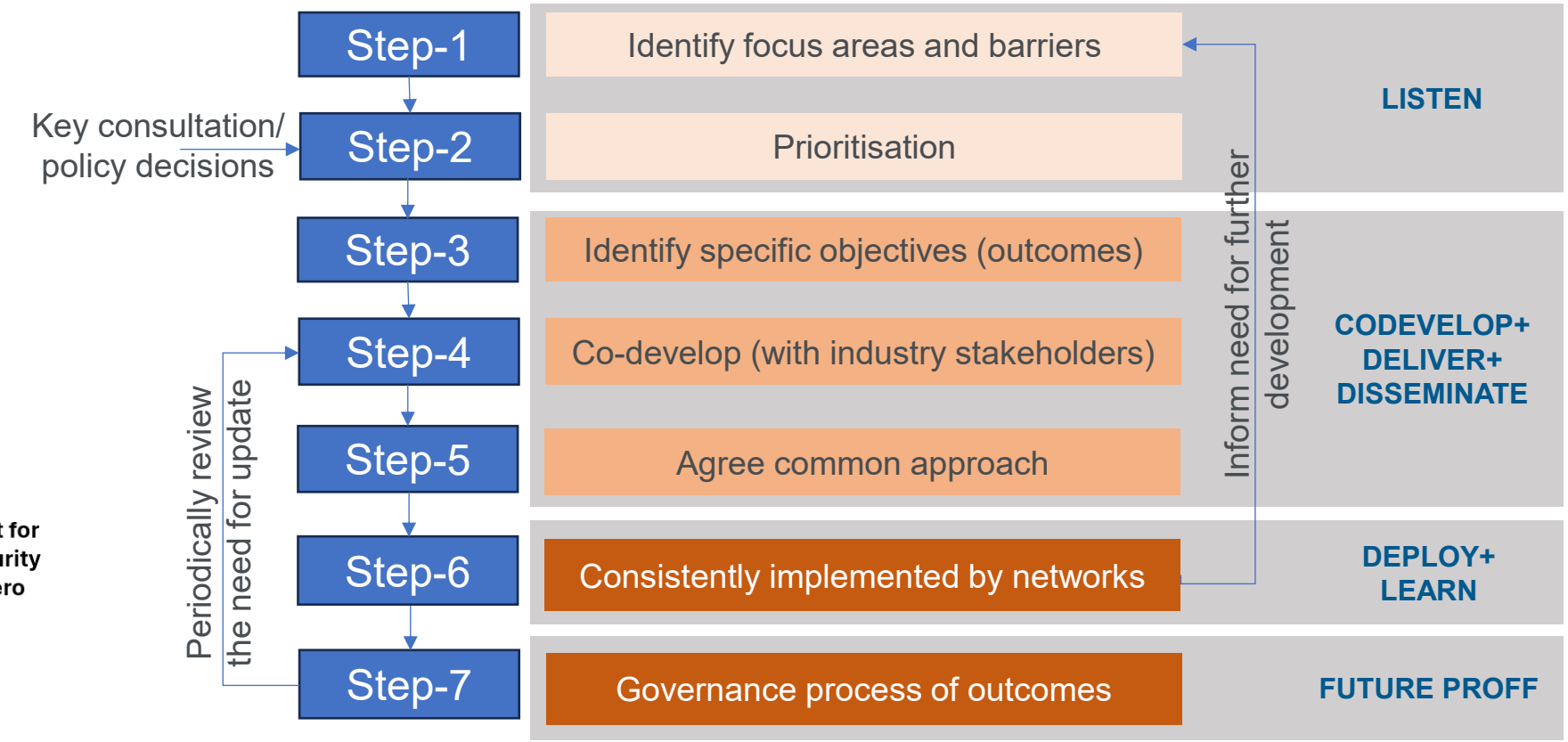




Flex contracted by networks has risen to **9GW roughly the capacity equivalent to 3 Hinckly Point C power stations**. A total of **22 GWh** of flexible energy was harnessed across the network – enough to power almost **7,000 average UK households for a full year**.

Open Networks - Our delivery approach





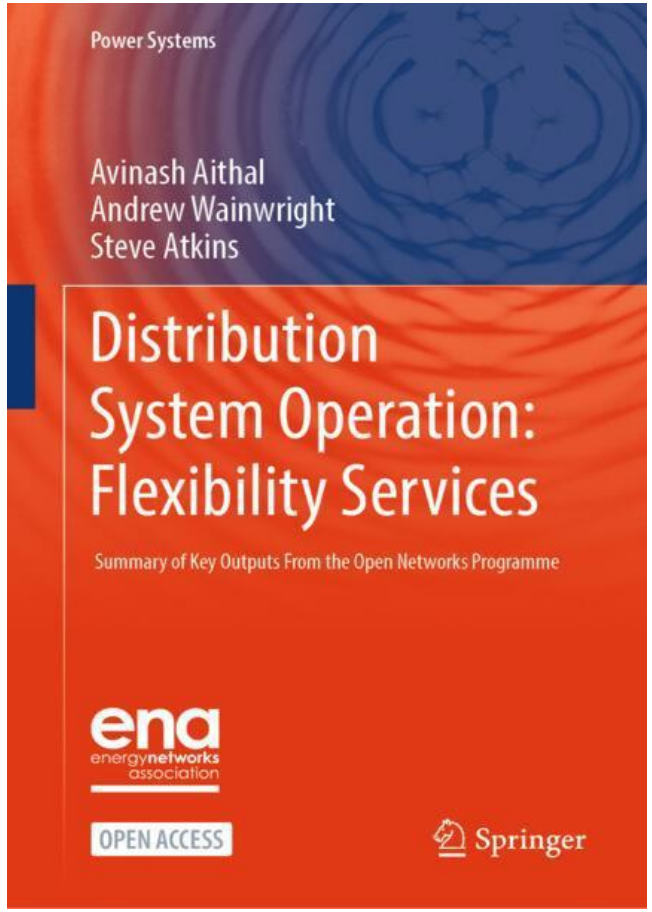
Delivered through technical working groups, that have **200 subject matter experts** from 8 network companies, supported by over **400 wider industry stakeholders**. and over **100 national and international delegates** feeding into the development of the programme

Key Learning from the Open Networks Programme

- **Summary**
  - It's important to understand the different types of flexibility and their business case.
  - Flexibility is a means to optimise network capacity and cannot be used to defer reinforcement indefinitely.
  - There is no market without market participants. It is important to engage market participants early in the journey in parallel to the establishment of institutional arrangements.
  - It is important to achieve a pragmatic balance between innovation and standardisation.

OPEN ACCESS

(Access via Link or QR code)



Speaker Biography

Dr Avinash Aithal

“ **Dr Avinash (Avi) Aithal** the Strategic Programme Director at Energy Networks Association (ENA) and formerly the Head of the Open Networks Programme. Avi is also a member of UKRI’s Science Engineering and Technology board, utilizing his expertise to champion multidisciplinary and emerging scientific research, laying the foundations of a smart grid in Great Britain.

Avi is a visionary clean energy and power systems leader with nearly two decades of experience in the energy sector, working in academia, policy delivery, business transformation, industrial research and manufacturing.

Avi is a trusted expert advising on national and international energy policy, a mentor, and a widely published author across academic, industry, and thought-leadership platforms.

Avi is a Chartered Engineer and a Fellow of the IET with a Masters in Electrical Energy Systems and a PhD in Smart Grids, both from Cardiff University, UK.””

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For discussions/suggestions/queries email: isuw@isuw.in

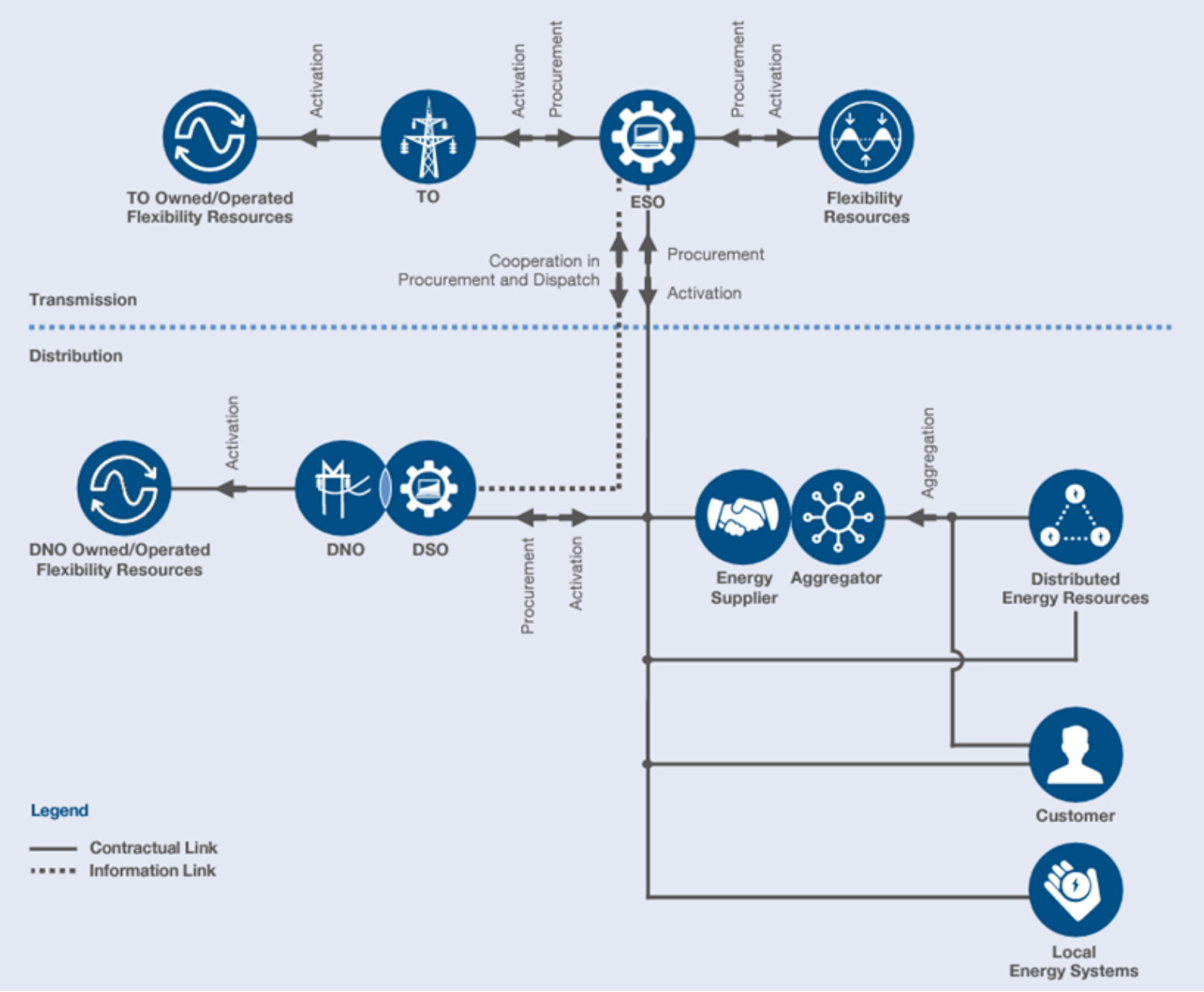
www.isuw.in

Links/References (If any)

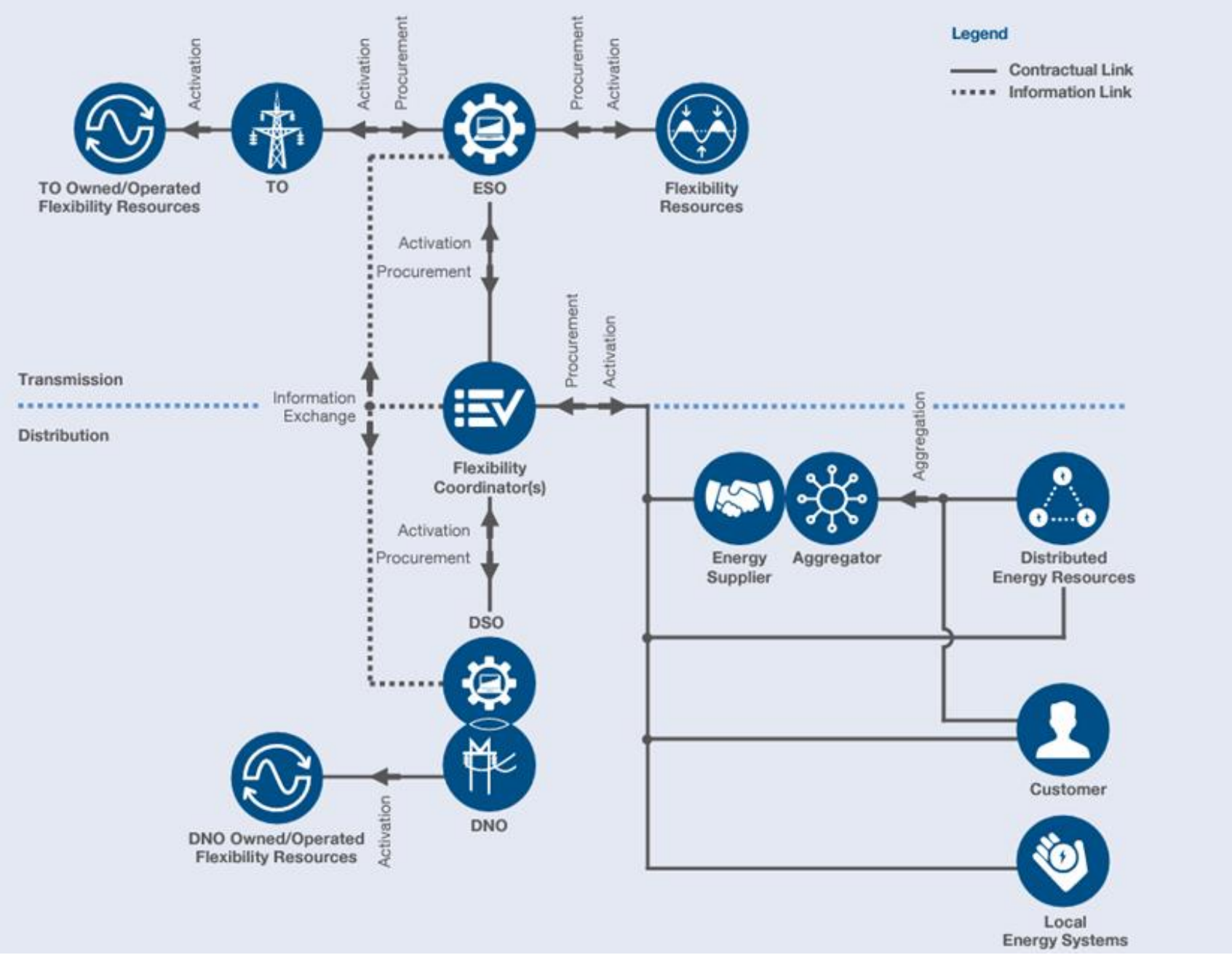
Back up slide

Existing "world"

Flexibility market arrangements (World-B)- Coordinated DSO–ESO Procurement & Dispatch



Back up slide

New “world”

Flexibility market arrangements (World E)- Flexibility Coordinator (Market Facilitator)